

VASCO60 Pipeline Check: Third-party POSS-I Candidate List Crossmatch

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Dataset: Third-party POSS-I candidate list (107,875 candidates) (not redistributed)

Vasco60 pipeline: <https://github.com/jannefi/vasco60>

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Main Conclusion

The dataset as-obtained is not a strict instantiation of the stated gates (as evaluated under the VASCO60 implementation of those gates). It contains systematic contaminants that could bias ensemble inference unless explicitly controlled.

This is supported by the numbers below. Of the candidates falling within the pipeline's sky footprint, 32.8% fail at least one quality gate when evaluated against our SExtractor photometry. The dominant failure — accounting for 94% of gate failures — is the SPREAD_MODEL morphology discriminator, indicating systematic contamination by extended sources (galaxies, blended objects) rather than point-like stellar candidates. A further 43.7% of in-footprint candidates pass the gates but are eliminated by subsequent pipeline stages, predominantly the shape and morphology stages (SOS, SOM).

1. Sky Coverage

	Count	% of dataset total
Dataset candidates (total)	107,875	100%
Candidates with tile coverage (in-footprint)	24,840	23.0%
Candidates with no tile coverage (out-of-footprint)	83,035	77.0%

Why are 77% out-of-footprint?

The pipeline uses only the **central, astrometrically reliable region** of each POSS-I plate. Tile centers are restricted to within 2.2° of each plate center, because the outer ring of every POSS-I plate exhibits strong geometric distortions and vignetting. Each plate subtends $\sim 6.5^\circ \times 6.5^\circ$ on sky, so this constraint deliberately excludes the outer portion of every plate.

Of the 83,035 out-of-footprint candidates:

Distance from nearest plate center	Count	% of uncovered
$\leq 2.2^\circ$ (in reliable zone but in inter-tile gap)	7,223	8.7%
$2.2^\circ - 3.25^\circ$ (plate-edge exclusion zone)	$\sim 50,800$	$\sim 61\%$
$3.25^\circ - 4.0^\circ$ (outside plate footprint)	$\sim 23,400$	$\sim 28\%$
$> 4.0^\circ$ (inter-plate gap)	~ 960	$\sim 1\%$

Note: Counts in this distance breakdown are rounded ($\sim$) because they come from binned distances.

$\sim 91\%$ of uncovered candidates fall in the plate-edge exclusion zone by design. These sit in the astrometrically unreliable outer ring of their plate and cannot be processed reliably. The remaining $\sim 8.7\%$ fall within the reliable central circle but between discrete 1° -spaced tile positions (inter-tile gaps inherent to the grid).

The $\sim 107\text{K}$ dataset was compiled from the full POSS-I plate area. VASCO60 pipeline covers only the astrometrically reliable central regions. This is a known, intentional footprint mismatch.

2. In-Footprint Outcome Funnel

All percentages below are relative to the 25,239 (candidate × tile) pairs — some candidates appear in two tiles due to border proximity, which is why pairs (25,239) slightly exceed unique candidates (24,840).

Outcome	Pairs	%
NO_MATCH: not detected by SExtractor	5,565	22.0%
GATE_FAIL: detected but fails quality gate(s)	8,286	32.8%
STAGE_ELIM: passes gates, eliminated by pipeline	11,027	43.7%
SURVIVOR: reaches final candidate list	361	1.4%

3. NO_MATCH Detail

Of 5,565 unmatched candidates:

Sub-reason	Count	% of NO_MATCH
circle_cut: detected but outside $\leq 30'$ inscribed circle	5,201	93.5%
not_detected: absent from SExtractor pass-2 catalog	364	6.5%

The circle cut is applied when building the pipeline's primary source list: only sources within a 30' radius of the tile center are retained, avoiding the tile corners where plate-edge effects begin to appear even within the reliable central area. The 5,201 candidates removed here were detected by SExtractor but fell in the corner zones of their tile. They are not pipeline survivors by geometry, not by photometric quality.

A supplementary check was performed against the **pass-1 SExtractor detection catalog** (cold-detection pass, prior to PSFEx modelling) for all 364 not_detected candidates. Each was projected onto the tile FITS image via WCS and crossmatched within 5" against the pass-1 LDAC catalog.

Pass-1 sub-classification	Count	% of not_detected
out_of_image: projects outside 60'x60' image bounds	190	52.2%
genuinely_undetected: interior, ≥ 50 px from edge, absent from pass-1	128	35.2%
outside_30' circle: in image corners (30'–42' from centre)	23	6.3%
edge_proximity: within 50 px ($\sim 85''$) of image boundary	15	4.1%
in_pass1: found in pass-1 within 5", absent from pass-2	8	2.2%

The dominant group (190) is a tile-matching artefact: assigned to a tile by angular proximity but projecting fractionally outside the actual FITS image footprint — never observable in this tile. The 128 **genuinely_undetected** cases are the scientifically meaningful figure: they fall well inside the image, inside the 30' pipeline circle, and are absent from SExtractor at both detection and measurement passes — consistent with the source being absent from the POSS-I image entirely or below the detection threshold. The 8 **in_pass1** cases were detected in the cold-detection pass but not promoted to pass-2 forced-position measurements, likely marginal detections rejected by PSFEx quality criteria.

4. Gate Failure Breakdown

8,286 candidates were detected and matched within 5", but fail at least one quality gate. A single candidate may fail multiple gates; the counts below are per-gate.

Gate	Failing candidates	% of GATE_FAIL pairs
SPREAD_MODEL > -0.002	7,779	93.9%
ELONGATION < 1.3	639	7.7%
SNR_WIN > 30	260	3.1%
FLAGS == 0	58	0.7%
2 < FWHM < 7	25	0.3%

SPREAD_MODEL is the dominant gate failure by a wide margin. SPREAD_MODEL measures the difference between the source profile and a PSF model; values below -0.002 indicate a source more compact than the PSF in the POSS-I image, consistent with a point source. Candidates failing this threshold appear as extended or resolved objects in our photometry — galaxies, nebulae, diffuse features, or blends.

This is a strong signal. Nearly one-third of all in-footprint candidates have SPREAD_MODEL morphology inconsistent with a stellar point source. The ELONGATION failures (639) provide secondary evidence of non-stellar or blended morphology.

5. Pipeline Stage Elimination

Of the 11,027 candidates that pass all quality gates, breakdown by eliminating stage:

Stage	Eliminated	% of STAGE_ELIM
S0S — Shape filter	5,161	46.8%
PRE_S0 — Circle / plate-edge veto (pre-pipeline)	3,142	28.5%
S0M — Morphology filter	2,628	23.8%
S3 — SCOS (scan artifact veto)	51	0.5%
S1 — GSC catalogue crossmatch	33	0.3%
S4 — PTF crossmatch	12	0.1%

S0S (shape) and S0M (morphology) together account for 70.6% of all stage eliminations. The algorithms implemented in these stages follow the approach described in Busko et al. (arXiv:2603.20407). Combined with the SPREAD_MODEL gate failures, the pipeline's morphological screening at every level consistently flags the majority of dataset candidates as non-stellar in character.

PRE_S0 (3,142) represents candidates that pass all gates and are detected in the SExtractor catalog, but are subsequently excluded by the $\leq 30'$ circle constraint or plate-edge veto when building the pipeline's primary source list — a geometry effect, not a quality failure.

Final survivors: 361 candidates from the dataset reach the final pipeline output.

6. Summary Table

Category	Count	% of in-footprint pairs
In-footprint pairs checked	25,239	—
Not detected by SExtractor	5,565	22.0%
Detected, fails morphology gate (SPREAD_MODEL)	7,779	30.8%
Detected, fails other gate(s)	507	2.0%
Gate-passing, eliminated by shape/morphology stages (S0S+S0M)	7,789	30.9%
Gate-passing, eliminated by catalogue vetoes (S1/S2/S3/S4/S5)	96	0.4%
Gate-passing, geometry veto (PRE_S0/circle)	3,142	12.4%
Survivors	361	1.4%

Notes on Interpretation

- All gate thresholds are those stated in Solano et al. (2022), MNRAS 515(1):1380, and applied without modification.
- SExtractor parameters used are the pipeline defaults (pass-2 catalog, 5'' matching radius).
- The check is conditional on tile coverage. The 83,035 out-of-footprint candidates are excluded by sky geometry, not by any quality assessment, and their properties are unknown to this pipeline.
- The not_detected sub-class (364 pairs, 1.4% of in-footprint total) was fully crossmatched against pass-1 detection catalogs; 128 are confirmed absent from SExtractor at both passes.
- Whether the SPREAD_MODEL violations represent true extended-source contamination or a systematic POSS-I plate calibration effect requires further investigation. The statistical signal is unambiguous. Its physical interpretation is open.